torch.nn.utils.skip_init#

https://pytorch.org/docs/stable/generated/torch.nn.utils.skip_init.html

Description:

Given a module class object and args / kwargs, instantiate the module without initializing parameters / buffers. This can be useful if initialization is slow or if custom initialization will be performed, making the default initialization unnecessary. There are some caveats to this, due to the way this function is implemented: 1. The module must accept a device arg in its constructor that is passed to any parameters or buffers created during construction. 2. The module must not perform any computation on parameters in its constructor except initialization (i.e. functions from torch.nn.init). If these conditions are satisfied, the module can be instantiated with parameter / buffer values uninitialized, as if having been created using torch.empty().

Function Signature

```
torch.nn.utils.skip_init(module_cls, *args, **kwargs)
[source]#

Parameters

Returns
```

Returns:

Instantiated module with uninitialized parameters / buffers

Code Examples

```
>>> import torch
>>> m = torch.nn.utils.skip_init(torch.nn.Linear, 5, 1)
>>> m.weight
Parameter containing:
tensor([[0.0000e+00, 1.5846e+29, 7.8307e+00, 2.5250e-29,
1.1210e-44]],
       requires_grad=True)
>>> m2 = torch.nn.utils.skip_init(torch.nn.Linear,
in_features=6, out_features=1)
>>> m2.weight
Parameter containing:
tensor([[-1.4677e+24,  4.5915e-41,  1.4013e-45,  0.0000e+00,
-1.4677e+24,
          4.5915e-41]], requires_grad=True)
```

Detailed Documentation

Documentation

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1. The module must accept a device arg in its constructor that is passed to any parameters or buffers created during construction.

2. The module must not perform any computation on parameters in its constructor except initialization (i.e. functions from torch.nn.init).

If these conditions are satisfied, the module can be instantiated with parameter / buffer values uninitialized, as if having been created using torch.empty().

module_cls – Class object; should be a subclass of torch.nn.Module

args – args to pass to the module’s constructor

kwargs – kwargs to pass to the module’s constructor

Instantiated module with uninitialized parameters / buffers